

Performance Improvement using Dynamic Contention Window Adjustment for Initial Ranging in IEEE 802.16 P2MP Networks

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Abstract*—In IEEE 802.16 networks, initial ranging is a primary and important procedure of connection setup between subscriber stations and base station. The mandatory method defined in the standard of contention resolution is based on a truncated binary exponential backoff, with a fixed initial contention window size. However, the original algorithm neglects the possibility that the number of actively contending stations may change dynamically over time, leading to dynamically changing contention intensity. The major contribution of this paper is twofold: 1) we propose an accurate analytical model to analyze the performance of initial ranging requests in IEEE 802.16 networks. Two metrics, connection probability and average connection delay, are investigated to evaluate the network performance; 2) based on the above analysis, we propose an efficient performance improvement method by using dynamic window adjustment for initial ranging. Unlike the standard algorithm, this algorithm automatically adjusts the initial contention window to an optimal trade-off point between connection probability and connection delay. The performance reveals that improving the service capacity and buffer size of base station can optimize the connection probability and the average connection delay. The numerical results also show that the optimal contention window adjustment outperforms the algorithm in the standard.

Keywords— *Optimal contention window adjustment; IEEE 802.16; Markov chain; Binary exponential backoff.*

I. INTRODUCTION

Recently, more and more interests have been involved in the design and analysis of wireless access networks for metropolitan area communication [2, 3]. Study group 802.16 was formed under IEEE Project 802 to recommend an air interface specification for Wireless Metropolitan Area Networks (WMANs) [1]. This standard specifies the air interface of fixed broadband wireless access (BWA) systems supporting multimedia services. IEEE 802.16 aims to provide high-speed wireless Internet access similar to wired access technologies [2].

In IEEE 802.16 protocol stack, the medium access control layer (MAC) supports a primarily Point-to-MultiPoint (P2MP) architecture, with an optional mesh topology. The 802.16 MAC is connection-oriented. All traffic, including inherently connectionless traffic, is mapped into a connection. This provides the ability to request bandwidth with its associated QoS and traffic parameters for every connection. Each connection is associated with a service flow (SF) which is a unidirectional flow of MAC packets on a connection that is

provided with a particular QoS. The service flow defines the QoS parameters for the packets that are transmitted on this connection [1].

IEEE 802.16 P2MP architecture consists of two kinds of fixed (non-mobile) stations: subscriber stations (SSs) and base station (BS). The BS regulates all the communication in the network, i.e., there is no peer-to-peer communication directly between the SSs. Each SS can deliver voice and data using common interfaces [6]. The communication path between SSs and BS has two directions: uplink channel (from SSs to the BS) and downlink channel (from BS to SSs). The downlink channel is a broadcast channel, while the uplink channel is a shared channel by the SSs [6]. Both uplink and downlink can operate in different frequencies using Frequency Division Duplexing (FDD) or share the same frequencies using Time Division Duplexing (TDD). In both FDD and TDD systems, the uplink and downlink channels are structured into frames. In TDD, the frame is divided into two subframes, uplink and downlink subframes, where the uplink subframe follows the downlink subframe. The uplink subframe includes three periods: Initial Maintenance period, Request Connection Opportunities period, and Scheduled Data grants period. The BS announces these periods and associated burst classes in the preceding downlink subframes's UL-MAP. The BS can specify such periods in any order and length, and can group the periods allocated to the Initial Maintenance and Request Contention Opportunities and leave the remaining periods for data transmission.

Initial ranging is the primary and part of the admission control procedure. It is the process of acquiring the correct timing offset and power adjustments such that the SSs can successfully transmit. Upon power up, an SS first has to synchronize to the BS's clock which involves two procedures: registration and ranging. The BS periodically reserves bandwidth in the upstream channel for new SSs to register. The time for the mini-slots that are allocated for registration purpose is broadcasted in the downstream channel for new SSs to join the network. The ranging mechanism is then used to determine the round-trip delay (RTD) from each SS to the BS after registration has been carried out.

In this paper, we propose an accurate analytical model to analyze the performance of initial ranging requests. The model is built based on a three dimensional Markov chain with the assumption that the ranging request arrival and connection durations are according to Poisson and exponential distributions respectively. Two metrics, connection probability and average connection delay, are investigated to evaluate the network performance. Based on above analysis, we propose

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an efficient performance improvement method by using dynamic window adjustment for initial ranging. Unlike the standard algorithm, this algorithm automatically adjusts the initial contention window to an optimal trade-off point between connection probability and connection delay according to the traffic activity. Compared with the algorithm in the standard, the dynamic connection window adjustment scheme is effective not only when there are many active stations that contend with each other for channel access, but also when there are few stations. Furthermore, it does not need online computations.

II. PERFORMANCE ANALYSIS

In this section, we integrate typical queuing theory and characteristic of contention resolution to study the initial ranging of the networks with a Markov model, and give the expressions of connection probability and connection delay as functions of the contention window size.

A. Assumptions

In the initial ranging period, new stations may join the network. Since several SSs can access the channel simultaneously, collision may occur in this period. Without loss of generality, the following assumptions are used in our analytical model.

- 1) The BS service capacity is m , i.e., the BS can accept at most m ranging connection requests simultaneously. When a new ranging request arrives, if the connection channel is fully occupied and the total number of ranging request is more than $m+2$, then the new ranging request must enter into the contention period.
- 2) The buffer size of the BS is N . That is, when a new ranging request arrives, if there are N ranging requests entering the contention resolution process but not yet given channel resources, the new ranging request will be rejected. Without loss of generality, we assume $N > m$.
- 3) The arrival of ranging requests follows Poisson process with rate λ , and ranging request connection duration is exponential distributed with mean $1/\mu$. Let $\rho = \frac{\lambda}{m\mu} < 1$.
- 4) The collision probability of a ranging request is independent of the number of retry times (if any) but is dependent on the number of contention ranging requests.
- 5) The contention window size is dependent on the number of contention ranging request. The BS dynamic adjusts the window size and broadcast it to all the SSs. On the downlink (from BS to SS), the messages including contention window size are broadcasted to all SSs and an SS only picks up the messages destined to it.

B. Modeling

With the above assumptions, the connection activities in initial ranging can be modeled by a 3-dimensional Markov Chain $\{s(t), b(t), l(t)\}$, where $s(t)$ is the stochastic process representing the back-off stage $(0, \dots, r)$ at time t and it records the number of retries a ranging request suffered; $b(t)$ is the

stochastic process representing the back-off counter for a given ranging request at time t and records the number of transmission opportunities that the request shall defer before transmitting; $l(t)$ is the number of contending ranging requests at time t .

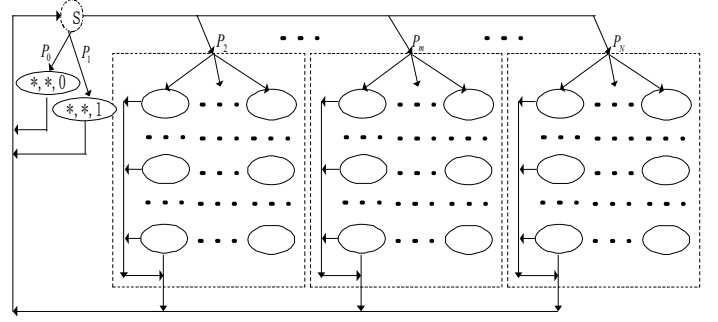


Figure 1. Overall Markov chain model for the ranging connections

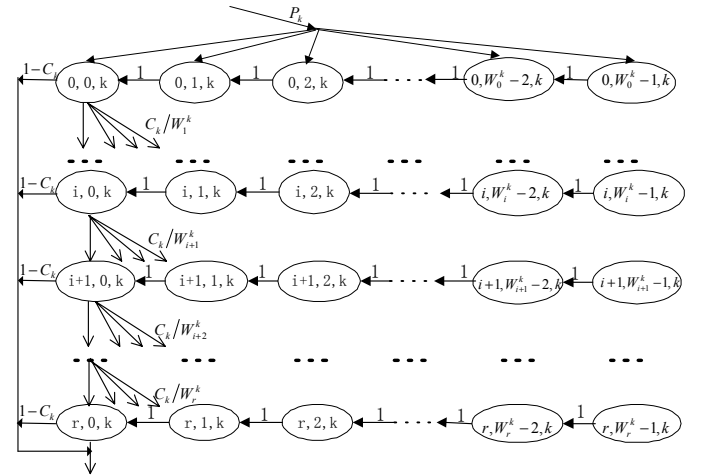


Figure 2. M_k - Markov chain model for the backoff window size for k ranging requests

With the assumptions 1), 2) and 3), the arrival of ranging request can be modeled as $M/M/m/N$ systems. According to $M/M/m/N$ model [4], we have the following formula which is derived from the steady-state probability transition diagram of $M/M/m/N$ system.

$$P_k = \begin{cases} P_0 \cdot \frac{(m\rho)^k}{k!} & k \leq m \\ P_0 \cdot \frac{m^m \rho^k}{m!} & m < k \leq N \end{cases} \quad (1)$$

Where P_0 is the probability that when a new ranging request arrives, there is idle channel to serve the ranging request and it can be established immediately. P_1 denotes the probability that when a new ranging request arrives, the channel is busy and the new ranging request has to wait for the service but there is no other ranging request contenting for the channel. P_i denotes the probability that when a new ranging request arrives, the channel is busy and there are $i-1$ ranging requests contenting the channel. Therefore the new ranging request must enter into the binary exponential back-off process.

Figure 1 shows the overall Markov chain of the ranging requests. When a new ranging request arrives, it will encounter the competitors and the corresponding probability

can be determined by (1). Note that the state s in Figure 1 is a pseudo-state and it is introduced for the convenience of presentation. Figure 2 shows the detailed Markov chain for k competitors (denoted as M_k , including the new connection) where the collision probability C_k depends on the number of competitors k , and is assumed to be a constant. W^k is the contention window size which is assigned by BS when there are k contention ranging request. W_i^k means the contention window size in stage i when the number of contention ranging request is k .

Based on Figure 1 and Figure 2, the one-step transition probabilities in the Markov chain are

$$\begin{cases} P\{i, j, k | i, j+1, k\} = 1 & i \in [0, r], j \in [0, W_i^k - 2], k \in [2, N] \\ P\{i, j, k | i-1, 0, k\} = \frac{C_k}{W_i^k} & i \in [1, r], j \in [0, W_i^k - 1], k \in [2, N] \\ P\{s | i, 0, k\} = C_k & i \in [0, r-1], k \in [2, N] \\ P\{s | r, 0, k\} = 1 & k \in [2, N] \\ P\{0, j, k | s\} = P_k & j \in [0, W_0 - 1], k \in [2, N] \\ P\{*, *, 0 | s\} = P_0 \\ P\{*, *, 1 | s\} = P_1 \end{cases} \quad (2)$$

The first equation in (2) accounts for the fact that when the SS receives a transmission opportunity, the backoff time counter is decremented. The second equation models the system after an unsuccessful ranging request. When an unsuccessful ranging request occurs at backoff stage $i-1$, the backoff stage increases and the new backoff value is uniformly chosen in the range $[0, W_i^k - 1]$. The third and fourth equations account for the fact that once the backoff stage reaches the maximum backoff stage r , after a successful ranging request or the rejection of the ranging request, the system is waiting for a new ranging request (return to state s). P_k denotes the probability of k ranging request competitors. The last two equations account for the fact that when the new ranging request arrives, the channel is not fully occupied or even the channel is busy, there are no other ranging competitors to contend for the channel.

Let $b_{i,j,k} = \lim_{t \rightarrow \infty} P\{s(t) = i, b(t) = j, l(t) = k\}$, $i \in [0, r]$,

$j \in [0, W_{i-1}^k]$, $k \in [2, N]$ be the stationary distribution of the Markov chain. We can immediately achieve

$$b_{0,0,k} = P_k \quad (3)$$

By using local balance equation for each stage in Fig. 1 and Figure 2, the probability of ranging request in state $\{i+1, 0, k\}$ can be deduced from the probability of ranging request of state $\{i, 0, k\}$ as below:

$$b_{i+1,0,k} = C_k \cdot b_{i,0,k} \quad 0 \leq i < r \quad (4)$$

Thus, (4) yields $b_{i,0,k} = C_k^i \cdot b_{0,0,k} \quad 0 < i \leq r$

Due to the regularities of the chain, for each $j \in [0, W_i^k - 1]$, $b_{i,j,k}$ becomes

$$b_{i,j,k} = \frac{W_i^k - j}{W_i^k} \cdot \begin{cases} P_k & i = 0 \\ C_k \cdot b_{i-1,0,k} & 0 < i \leq r \end{cases} \quad (5)$$

By means of relations (3) and (4), (5) can be rewritten as

$$b_{i,j,k} = \frac{W_i^k - j}{W_i^k} b_{i,0,k} \quad i \in [0, r] \quad j \in [0, W_i^k - 1] \quad (6)$$

Thus, by relations (1) (4) and (6), all the values $b_{i,j,k}$ are expressed as functions of the value P_0 and the collision probability C_k . Applying the normalization condition, we have

$$\begin{aligned} 1 &= P_0 + P_1 + \sum_{k=2}^N \sum_{i=0}^r \sum_{j=0}^{W_i^k-1} b_{i,j,k} \\ &= P_0 + m\rho P_0 + \sum_{k=2}^m P_0 \frac{(m\rho)^k}{k!} \sum_{i=0}^r b_{i,0,k} \sum_{j=0}^{W_i^k-1} \frac{W_i^k - j}{W_i^k} + \sum_{k=m+1}^N P_0 \frac{m^m \rho^k}{m!} \sum_{i=0}^r b_{i,0,k} \sum_{j=0}^{W_i^k-1} \frac{W_i^k - j}{W_i^k} \\ &= P_0 + m\rho P_0 + \sum_{k=2}^m P_0 \frac{(m\rho)^k}{k!} \sum_{i=0}^r b_{i,0,k} \frac{W_i^k + 1}{2} + \sum_{k=m+1}^N P_0 \frac{m^m \rho^k}{m!} \sum_{i=0}^r b_{i,0,k} \frac{W_i^k + 1}{2} \\ &= P_0 + m\rho P_0 + \sum_{k=2}^m P_0 \frac{(m\rho)^k}{k!} \sum_{i=0}^r C_k \frac{W_i^k + 1}{2} + \sum_{k=m+1}^N P_0 \frac{m^m \rho^k}{m!} \sum_{i=0}^r C_k \frac{W_i^k + 1}{2} \\ &= P_0 \left[1 + m\rho + \sum_{k=2}^m \frac{(m\rho)^k}{k!} \sum_{i=0}^r C_k \frac{W_i^k + 1}{2} + \sum_{k=m+1}^N \frac{m^m \rho^k}{m!} \sum_{i=0}^r C_k \frac{W_i^k + 1}{2} \right] \end{aligned} \quad (7)$$

Once we derive P_0 and C_k , the stationary probability of any state (i, j, k) can be calculated by the above equations. We can now express τ_k , which denotes the probability that the ranging request in a randomly chosen slot time contending with $k-1$ other ranging competitors. As any initial ranging transmission occurs when the backoff time counter is equal to zero, regardless of the backoff stage, it is

$$\begin{aligned} \tau_k &= \sum_{i=0}^r C_k \cdot b_{i,0,k} = \sum_{i=0}^r C_k^i b_{0,0,k} = \sum_{i=0}^r C_k^i \cdot \begin{cases} \frac{(m\rho)^k}{k!} P_0 & k \leq m \\ \frac{m^m \rho^k}{m!} P_0 & m < k \leq N \end{cases} \quad (8) \\ &= \begin{cases} \frac{1 - C_k^{m+1}}{1 - C_k} \cdot \frac{(m\rho)^k}{k!} P_0 & k \leq m \\ \frac{1 - C_k^{m+1}}{1 - C_k} \cdot \frac{m^m \rho^k}{m!} P_0 & m < k \leq N \end{cases} \quad 2 \leq k \leq N \end{aligned}$$

In (8), τ_k depends on the conditional collision probability C_k , which is still unknown. It is sufficient to note that C_k is the probability that, in a time slot, at least one of the $k-1$ remaining connection requests is trying to access the channel. At steady state, each remaining ranging request accesses to the channel with probability C_k . This yields

$$C_k = 1 - (1 - \tau_k)^{k-1} \quad 2 \leq k \leq N \quad (9)$$

It can be seen that (7) (8) and (9) form a nonlinear equation systems with unknown variable P_0 , τ_k and C_k ($i=2, \dots, N$), which can be solved by numerical method.

C. Analyses of connection probability and connection delay

Let F be the connection probability of the network. It is defined as the probability that a ranging request is accepted in a given network. The unsuccessful connection probability arises for an incoming ranging request due to two situations: 1) system busy probability (UP_1) when there are N ranging requests waiting in the queue of BS and the channel is busy at

the moment, the ranging request is unable to enter into the network and gets lost; and 2) Max backoff stage probability (UP_2) when the ranging request is in the process of binary exponential backoff and reaches the maximum stage r , it will be rejected if it still collides with other ranging requests. Since P_0 can be solved in the nonlinear equation systems (7) (8) and (9) and UP_1 is equal to the probability that there are N ranging requests in the system, according to (1), we obtain

$$UP_1 = P_N = P_0 \cdot \frac{m^m \rho^N}{m!}$$

Still according to (1), solving for P_0 and C_k ($k=2, \dots, N$) in the nonlinear equation systems (7) (8) and (9), we obtain

$$UP_2 = \sum_{k=1}^N P_k \cdot C_k^{r+1} = \sum_{k=2}^m (P_0 \cdot \frac{(m\rho)^k}{k!} \cdot C_k^{r+1}) + \sum_{k=m+1}^N (P_0 \cdot \frac{m^m \rho^k}{m!} \cdot C_k^{r+1})$$

UP_1 and UP_2 are the reject probabilities of ranging request. By integrating the above two formulas, we obtain the connection probability as

$$F = 1 - (UP_1 + UP_2) = 1 - [P_0 \cdot \frac{m^m \rho^N}{m!} + \sum_{k=2}^m (P_0 \cdot \frac{(m\rho)^k}{k!} \cdot C_k^{r+1}) + \sum_{k=m+1}^N (P_0 \cdot \frac{m^m \rho^k}{m!} \cdot C_k^{r+1})] \quad (10)$$

Let D be the average connection delay for ranging requests, defined as average number of slots times that a ranging request suffers from entering the network to the establishment of the connection. In Fig. 2, for M_k , k ranging requests collide with probability C_k , and a connection succeeds until it receives an acknowledgement (thus indicating a successful connection), so we can model the number of connection times as geometrically distributed with success probability $1 - C_k$. Furthermore, each collision causes a dilation of the contention window until the maximum is reached, thus the expected backoff window size is

$$W_k^{backoff} = (1 - C_k) \frac{W^k}{2} + C_k (1 - C_k) \frac{2W^k}{2} + \dots + C_k^r (1 - C_k) \frac{2^r W^k}{2} \\ = \frac{(1 - C_k)(1 - (2C_k)^{r+1})}{1 - 2C_k} \cdot \frac{W^k}{2}$$

Therefore, D can be calculated according to the expected backoff window size for M_k

$$D = \sum_{k=2}^N P_k \cdot W_k^{backoff} = \sum_{k=2}^m \left[P_0 \cdot \frac{(m\rho)^k}{k!} \cdot \frac{(1 - C_k)(1 - (2C_k)^{r+1})}{1 - 2C_k} \cdot \frac{W^k}{2} \right] \\ + \sum_{k=m+1}^N \left[P_0 \cdot \frac{m^m \rho^k}{m!} \cdot \frac{(1 - C_k)(1 - (2C_k)^{r+1})}{1 - 2C_k} \cdot \frac{W^k}{2} \right]$$

III. PERFORMANCE IMPROVEMENT USING DYNAMIC WINDOW ADJUSTMENT

In this section, we propose a scheme to improve the performance of initial connection in IEEE 802.16 networks based on the above analysis.

A. Definition of the problem

Based on the performance analysis in section II.B, we have the following constrained conditions

$$\begin{cases} 1 = P_0 \left[1 + m\rho + \sum_{k=2}^m \frac{(m\rho)^k}{k!} \sum_{i=0}^r C_k^i \frac{W_i^k + 1}{2} + \sum_{k=m+1}^N P_0 \frac{m^m \rho^k}{m!} \sum_{i=0}^r C_k^i \frac{W_i^k + 1}{2} \right] \\ C_k = 1 - (1 - \tau_k)^{k-1} \\ \tau_k = \begin{cases} \frac{1 - C_k^{m+1}}{1 - C_k} \cdot \frac{(m\rho)^k}{k!} P_0 & k \leq m \\ \frac{1 - C_k^{m+1}}{1 - C_k} \cdot \frac{m^m \rho^k}{m!} P_0 & m < k \leq N \end{cases} \\ W^k > 0 \text{ and } W^k \text{ is an integer} \end{cases} \quad 2 \leq k \leq N \quad (11)$$

We can see that the constrained conditions are equation constrained conditions including the variables P_0, τ_k, C_k and W^k ($2 \leq k \leq N$). For any group of $W = \{W^k \mid 2 \leq k \leq N\}$, we can calculate the variables of P_0, τ_k, C_k from the nonlinear equations systems. In section II.C, the expression of connection probability and connection delay are dependent on P_0, C_k and W . We denote the constrained conditions of (11) as $G(W)=0$. The problem that we are concerned about can be defined as follows:

The performance improvement problem is to find a vector $W^* \in W$ that satisfies the following conditions:

- 1) The connection probability F is maximum
- 2) The connection delay D is minimum

In general, the two metrics can not be optimized simultaneously for $\forall W' \in W$.

B. Optimize the connection probability and connection delay

Two single objective programs were proposed to optimize connection probability and connection delay respectively.

- 1) Enhance the connection probability

$$\text{Maximize } F = 1 - [P_0 \cdot \frac{m^m \rho^N}{m!} + \sum_{k=2}^m (P_0 \cdot \frac{(m\rho)^k}{k!} \cdot C_k^{r+1}) + \sum_{k=m+1}^N (P_0 \cdot \frac{m^m \rho^k}{m!} \cdot C_k^{r+1})] \quad (12)$$

Subject to $G(W)=0$

- 2) Reduce the connection delay

$$\text{Minimize } D = \sum_{k=2}^m \left[P_0 \cdot \frac{(m\rho)^k}{k!} \cdot \frac{(1 - C_k)(1 - (2C_k)^{r+1})}{1 - 2C_k} \cdot \frac{W^k}{2} \right] \\ + \sum_{k=m+1}^N \left[P_0 \cdot \frac{m^m \rho^k}{m!} \cdot \frac{(1 - C_k)(1 - (2C_k)^{r+1})}{1 - 2C_k} \cdot \frac{W^k}{2} \right] \quad (13)$$

Subject to $G(W)=0$

Note that (12) and (13) are integer programming problems; branch-and-bound algorithm can be used to solve them.

C. Trade-off between connection probability and connection delay

As mentioned above, the connection probability and the connection delay can not be optimized simultaneously for $\forall W' \in W$. So there is a general equivalence between maximizing connection probability and minimizing

connection delay. In order to achieve the trade-off between connection probability and connection delay, we proposed a multiple objective program model.

$$V - \max H(W) = \{F, -D\}$$

Subject to

$$G(W) = 0 \quad (14)$$

Note that the second element in the objective function is $-D$.

To achieve the trade-off of the above multiple objective program, we define the optimized solution of (11) and (12) are F^* and D^* respectively. Denote $H^* = (F^*, -D^*)^T$, then H^* is called an ideal point. An efficient scheme is to find a point that is closed to the ideal point. That is

$$\|H - H^*\| \rightarrow \min \|H - H^*\| \quad (15)$$

This function is called utility function. In general, the p -norm of the utility function is

$$L_p(W) = \|H(W) - H^*\| = [(F - F^*)^p + (D - D^*)^p]^{1/p}$$

Hence the multiple objective programs can be changed to the following single objective program model.

$$\text{Minimize } L_p(W) = [(F - F^*)^p + (D - D^*)^p]^{1/p} \quad (16)$$

$$\text{Subject to } G(W) = 0$$

D. Dynamic contention window adjustment

The contention window adjustment algorithm defined in 802.16 has been proven to be the robust in theoretical analysis. However, with a fixed CW_{min} , the original algorithm neglects the possibility that the number of actively contenting stations can change dynamically over time, leading to dynamically changing contention intensity. When there are many active stations, a too small CW_{min} may lead to too many collisions and backoffs; on the other hand, when there are few active stations, too high a CW_{min} may lead to excessive idle time during which no station attempts to access the channel. In either case, the channel is not used efficiently.

To solve the problem, one could employ a dynamic contention window adjustment algorithm with contention window adjusted to reflect the contention intensity in the network. In the above analysis, the relationship between the trade-off of performance metrics and the optimal contention window is established. The computation can be calculated by BS in advance and the BS broadcasts the optimal contention window size to all SSs according to the number actively contending stations.

IV. EXPERIMENTAL EVALUATION

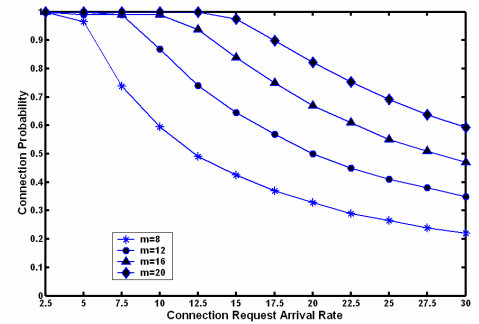
This section presents evaluation of the network performance. Our simulator is Mesquite CSIM C [5], a process-oriented, general purpose simulation toolkit. All the simulations run on SUN Ultra-SPARC workstations. We assume that the ranging requests for connection establishment form a Poisson process with λ , while the connection lifetimes are exponentially distributed with average of 10 seconds.

A. Effect of service capacity and buffer size of BS

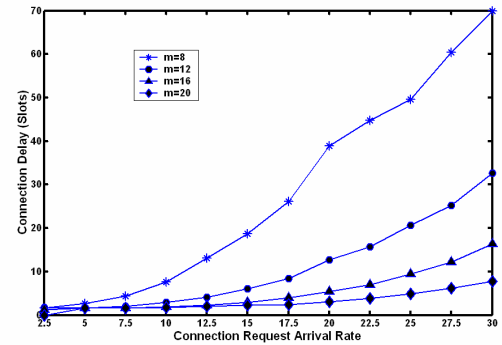
In this subsection, we report numerical results and make observations. The results are all obtained from analysis and

simulation. The symbols in the following figures represent the analytical results and the lines represent the simulation data. Due to limited space, we only present a limited number of simulation cases in this paper. However, we find that the conclusions we draw through these cases are generally true for many other cases we have evaluated. Note that the simulation results are the average of 100 runs with different seeds for random number generation.

Figure 3 shows the effect of the service capacity of the network to the connection probability and average connection delay under the parameters $\{W=32, r=5, N=100\}$. The arrival rate of ranging request ranges from 2.5 to 30.0. It is observed that the connection probability decreases with the increase of ranging request arrival rate while the average connection delay increases with the increase of ranging request arrival rate. However, when the service capacity of the network increases, the connection probability improves sharply and the connection delay drops quickly.

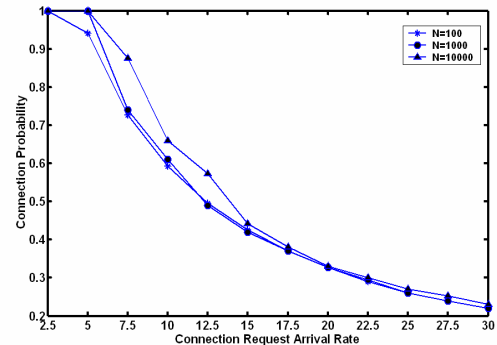


(a) Connection Probability



(b) Average Connection Delay

Figure 3. Effect of the service capacity to the two performance metrics



(a) Connection Probability

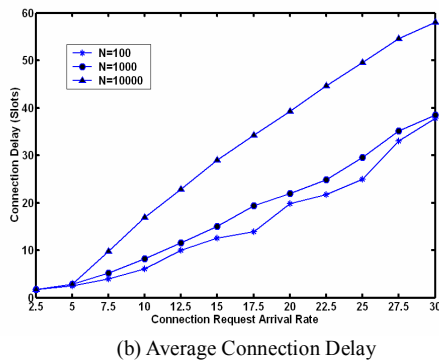


Figure 4. Effect of buffer size to the two performance metrics

Figure 4 shows the effect of buffer size to the connection probability and average connection delay under the parameters $\{W=32, r=5, m=15\}$. These figures also show that increasing the buffer size can improve the connection probability whereas increases average connection delay.

B. Numeric results of performance improvement

This subsection demonstrates the performance through applying our improvement method. For simplicity, we use $\langle A \rangle$ to represent the system setting, in which $A \in \{\text{CP-opt}, \text{CD-opt}, \text{CP-CD-tradeoff}, \text{Non-opt}\}$, where CP-opt means the optimize objective is connection probability, CD-opt means the optimize objective is connection delay, CP-CD-tradeoff means the trade-off between them. And Non-opt means that the BS broadcast the same window size $W=32$. The parameters setting is as follows: $m=10, N=100$ and $r=5$.

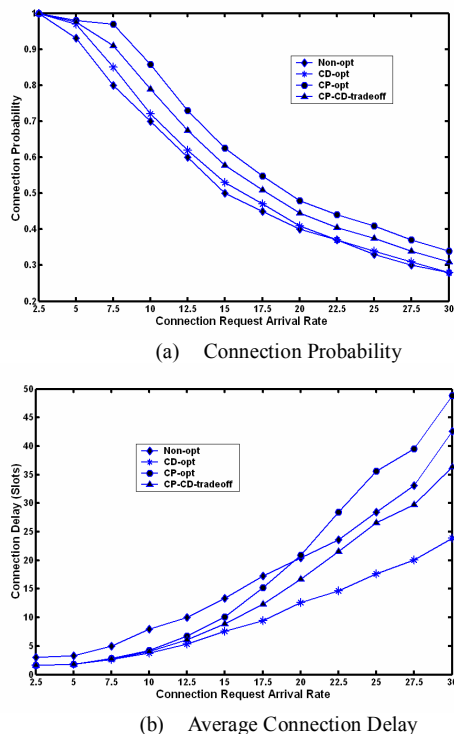


Figure 5. Effect of performance improvement method to the two metrics

Figure 5 shows the connection probability and connection delay vary with the connection request arrival rate. As expected, the connection probability decreases while the connection delay increases as the request arrival rate increases. The higher the request arrival rate increases, the more the number of contention connections. As the figures shown, the CP-opt and CD-opt can enhance the connection probability and reduce the connection delay than Non-opt. CP-opt has higher connection but also has higher connection delay probability than CD-opt. Fortunately, CP-CD-tradeoff can achieve relative high connection probability and low connection delay. This demonstrates that the dynamic contention window adjustment outperforms the algorithm in the standard.

V. CONCLUSIONS

In this paper, we propose an accurate analytical model to analyze the performance of initial ranging requests in IEEE 802.16 networks. The model is built based on a three dimensional Markov chain with the assumption that the ranging request arrival and connection durations are according to Poisson and exponential distributions respectively. Two metrics, connection probability and average connection delay, are investigated to evaluate the network performance. The performance reveals that improving the service capacity and buffer size of base station can optimize the connection probability and the average connection delay.

Based on the above analysis, we propose an efficient performance improvement method by using dynamic window adjustment for initial ranging. Unlike the standard algorithm, this algorithm automatically adjusts the initial contention window to an optimal trade-off point between connection probability and connection delay. The numerical results also show that the dynamic contention window adjustment outperforms the algorithm in the standard and it is effective and simple to implement.

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